Effect of density and magnetic field fluctuations on rf waves propagation

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1 Introduction

Low frequency fluctuations of the electron density or the magnetic field can strongly affect the propagation of rf waves in a magnetized plasma. This effect has been extensively investigated for the Lower Hybrid (LH) wave by solving a wave kinetic equation in the weak turbulence approximation [1, 2]. In this approach, the flow of rf energy is described by a usual ray-tracing between random scattering events where the initial wave vector $\mathbf{k}$ can be modified both in amplitude and direction with the constraint to remain solution of the local dispersion relation.

This physical mechanism is a good candidate for explaining a possible strong upshift of the power spectrum as the LH wave propagates in the plasma, even if the toroidal mode coupling for bridging the spectral gap is very weak. Though primarily dedicated to the LH wave, the concept of wave scattering by low frequency density or magnetic field fluctuations can be easily extended to other types of rf waves provided $\widetilde{k}_{\|} \ll k_{\|}$ and $\widetilde{\omega} \ll \omega$. Here, $k_{\|} = \mathbf{k} \cdot \widehat{\mathbf{b}}$ and $\widetilde{k}_{\|} = \widetilde{\mathbf{k}} \cdot \widehat{\mathbf{b}}$ are the projections of the rf and fluctuation wave vectors respectively along the local direction of the magnetic field $\widehat{\mathbf{b}} = \mathbf{B}/\|\mathbf{B}\|$, while ω and $\widetilde{\omega}$ are the rf and fluctuation frequencies respectively. Consequently, the scattering process becomes a two dimensional problem in the plane perpendicular to the local magnetic field direction, a reasonable assumption regarding the large difference

between the long wave length of the low frequency fluctuations of the plasma and the short length of the rf waves along $\widehat{\mathbf{b}}$.[1]

Calculations are organized in three steps. In the first section, the components of the scattered rf wave vector $\check{\mathbf{k}}$ are calculated, according to the curvilinear coordinate system chosen for describing the magnetic equilibrium in C3PO [5]. The wave kinetic equation corresponding to the wave-wave interaction is then derived from first principles in the second section for any kind of rf waves and plasma model. Finally, in the third section, the wave kinetic equation is solved using a Monte-Carlo method and the procedure for the numerical implementation in the code is explicited.

1.0.1 Calculation of the scattered wave vector

When the rf wave vector $\mathbf{k}$ is scattered by low density fluctuations, it may change both in module and direction, with the condition

$$\mathbf{k} \cdot \widehat{\mathbf{b}} = \check{\mathbf{k}} \cdot \widehat{\mathbf{b}} = k_{\parallel} \tag{1}$$

while $\check{\mathbf{k}}$ is still solution of the local dispersion relation. Consequently, the scattering process may be expressed in the simple form

$$\mathbf{k}_{\perp} \cdot \check{\mathbf{k}}_{\perp} = k_{\perp} \check{k}_{\perp} \cos\beta, \tag{2}$$

where $\check{k}_{\perp} = \left\|\check{\mathbf{k}}_{\perp}\right\| = \left\|\check{\mathbf{k}} \times \widehat{\mathbf{b}}\right\|$ may change by wave-wave interaction if several modes of propagation are solutions of the dispersion relation for the same $k_{\parallel}$, while β is the angle of the rotation, between the initial and scattered wave vector directions in the plane perpendicular to $\widehat{\mathbf{b}}$.[2] Unlike transitions are defined as those corresponding to $\check{k}_{\perp} \neq k_{\perp}$, while like ones to $\check{k}_{\perp} = k_{\perp}$.

Using the coordinate system defined in Ref. [5], the components $\left(\check{k}_{\rho}, \check{m}, \check{n}\right)$ of $\check{\mathbf{k}}$ are thus fully determined by the components (k_{ρ}, m, n) of the wave vector $\mathbf{k}$ prior the scattering event and the two parameters β and $\chi = \check{k}_{\perp}/k_{\perp}$.

The conservation (1) of the parallel component of the wave vector in the scattering process may be expressed in the form,

$$\sigma_I P \frac{\check{m}}{r} \cos\alpha + \sigma_B T \frac{\check{n}}{R} = \sigma_I P \frac{m}{r} \cos\alpha + \sigma_B T \frac{n}{R} \tag{3}$$

which gives the first equation which must be satisfied by components of $\check{\mathbf{k}}$ defined as

[1] The parallel wave length $\lambda_{\parallel}$ is related to the parallel refractive index $N_{\parallel}$ of the rf wave by the simple relation $\lambda_{\parallel} = c/fN_{\parallel}$,where $f = \omega/2\pi$ is the rf wave frequency. Using typical values corresponding to rf systems used on the Tore Supra tokamak [3, 4], $f_{LH} = 3.7GHz$ and $N_{\parallel LH} \simeq 2$, $\lambda_{\parallel LH}$ is about $4cm$ for the LH wave, while it is $\lambda_{\parallel EC} \simeq 1.3cm$ for the Electron Cyclotron (EC) wave with $f_{EC} = 110GHz$ and $N_{\parallel EC} \simeq 0.2$. In both cases, the condition of validity of the model $\lambda_{\parallel} \ll \widetilde{\lambda}_{\parallel}$ where $\widetilde{\lambda}_{\parallel} \simeq qR_p$ is well fullfiled. since the safety factor $q > 1$, and $R_p \sim m$.

[2] In Ref. [1], the relation $\widehat{\mathbf{b}} \cdot \left(\mathbf{k}_{\perp} \times \check{\mathbf{k}}\right) = k_{\perp} \check{k}_{\perp} \sin\beta$ is also introduced. However, it is fully equivalent to (2) and does not provide any additional information.

$$\check{\mathbf{k}} = \check{k}_\rho \left\|\nabla\rho\right\| \widehat{\rho} + \frac{\check{m}}{r}\widehat{\theta} + \frac{\check{n}}{R}\widehat{\phi} \tag{4}$$

according to [5]. Here, $\widehat{\mathbf{b}} = \sigma_B T\widehat{\phi} + \sigma_I P\hat{\mathbf{s}}$ with $T = B_T/B$ and $P = B_P/B$, where σ_B is the sign of B_ϕ and σ_I is the sign of I_ϕ. $B_T = \left|I\left(\psi\right)\right|/R$ and $B_P = \left\|\nabla\psi\right\|/R$ are definite positive and $\hat{\mathbf{s}} = \widehat{\phi}\times\widehat{\rho}$ with $\nabla\rho = \left\|\nabla\rho\right\|\widehat{\rho}$ such that $\hat{\mathbf{s}}$ is orientated counter-clockwise in the poloidal plane.

Expressing the scalar product $\mathbf{k}_\perp \cdot \check{\mathbf{k}}_\perp$ as

$$\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\cdot\left(\check{\mathbf{k}}\times\widehat{\mathbf{b}}\right) = \left(\mathbf{k}\cdot\check{\mathbf{k}}\right)\left(\widehat{\mathbf{b}}\cdot\widehat{\mathbf{b}}\right) - \left(\mathbf{k}\cdot\widehat{\mathbf{b}}\right)\left(\widehat{\mathbf{b}}\cdot\check{\mathbf{k}}\right) = \mathbf{k}\cdot\check{\mathbf{k}} - k_\parallel^2 \tag{5}$$

and using (4), (2) becomes

$$k_\perp\check{k}_\perp\cos\beta = k_\perp^2\chi\cos\beta = k_\rho\check{k}_\rho\left\|\nabla\rho\right\|^2 + \frac{m\check{m}}{r^2} + \frac{n\check{n}}{R^2} - \frac{\left\|\nabla\rho\right\|\sin\alpha}{r}\left(m\check{k}_\rho + \check{m}k_\rho\right) - k_\parallel^2 \tag{6}$$

since $\hat{\rho}\cdot\hat{\theta} = -\sin\alpha$ and $\hat{\theta}\cdot\hat{\phi} = \hat{\rho}\cdot\hat{\phi} = 0$ [5]. This the second equation between components of $\check{\mathbf{k}}$.

Since $\check{k}_\perp$ is determined from the dispersion relation from which $\check{k}_\perp$ is evaluated, the third equation is simply

$$\chi^2 k_\perp^2 = \check{k}_\rho^2\left\|\nabla\rho\right\|^2 - 2\check{k}_\rho\left\|\nabla\rho\right\|\frac{\check{m}\sin\alpha}{r} + \frac{\check{m}^2}{r^2} + \frac{\check{n}^2}{R^2} - k_\parallel^2 \tag{7}$$

The set of equations (3), (6) and (7) gives an implicit formulation of the components of $\check{\mathbf{k}}$ as a function of those of $\mathbf{k}$. However, even if $\check{k}_\rho$, $\check{m}$ and $\check{n}$ can be fully defined in this way, this formulation is not appropriate for fast numerical calculations, owing to the non-linearity of (7).

A more suitable approach for an efficient numerical implementation consists in deriving an explicit formulation of the components of $\check{\mathbf{k}}$ as a function of $\mathbf{k}$, β and χ, considering the geometrical transformation from $\mathbf{k}$ to $\check{\mathbf{k}}$. Let define the orthonormal frame of reference $\left(\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}/k_\perp, \mathbf{k}\times\widehat{\mathbf{b}}/k_\perp, \widehat{\mathbf{b}}\right)$ formed by the local directions $\widehat{\mathbf{b}}$ and $\mathbf{k}$ of the magnetic field and wave vector respectively. The scattered wave vector may be expressed

$$\begin{aligned}\check{\mathbf{k}} &= \left(\check{\mathbf{k}}\cdot\widehat{\mathbf{b}}\right)\widehat{\mathbf{b}} + \left(\check{\mathbf{k}}\cdot\frac{\mathbf{k}\times\widehat{\mathbf{b}}}{k_\perp}\right)\frac{\mathbf{k}\times\widehat{\mathbf{b}}}{k_\perp} + \left(\check{\mathbf{k}}\cdot\left[\frac{\mathbf{k}\times\widehat{\mathbf{b}}}{k_\perp}\times\widehat{\mathbf{b}}\right]\right)\left[\frac{\mathbf{k}\times\widehat{\mathbf{b}}}{k_\perp}\times\widehat{\mathbf{b}}\right] \\ &= \check{k}_\parallel\widehat{\mathbf{b}} + \check{k}_\perp\sin\beta\frac{\mathbf{k}\times\widehat{\mathbf{b}}}{k_\perp} - \check{k}_\perp\cos\beta\frac{\mathbf{k}\times\widehat{\mathbf{b}}}{k_\perp}\times\widehat{\mathbf{b}} \\ &= \check{k}_\parallel\widehat{\mathbf{b}} + \chi\left(\sin\beta\mathbf{k}\times\widehat{\mathbf{b}} - \cos\beta\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\right)\end{aligned} \tag{8}$$

where

$$\check{k}_\perp = \check{\mathbf{k}}\cdot\frac{\mathbf{k}\times\widehat{\mathbf{b}}}{k_\perp} \tag{9}$$

and it can be easily verified that for $\beta = 0$ and $\chi = 1$, $\check{\mathbf{k}} = \mathbf{k}$ is well recovered, since $\check{k}_{\|} = k_{\|}$ in the model. From (8), the second right handside term corresponds to a rotation of an angle β combined with an homothetic transformation of the module of the perpendicular component of the wave vector.

The scattering operator $\mathcal{S}_{\beta}^{\chi}(\mathbf{k})$ is then

$$\mathcal{S}_{\beta}^{\chi}(\mathbf{k}) = \left(\mathbf{k}\cdot\widehat{\mathbf{b}}\right)\widehat{\mathbf{b}} + \chi\left(\sin\beta\mathbf{k}\times\widehat{\mathbf{b}} - \cos\beta\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\right) \tag{10}$$

and $\mathcal{S}_0^1 = \mathcal{I}$, where $\mathcal{I}$ is the identity transformation.

From (4),

$$\check{k}_{\rho}\|\nabla\rho\| - \frac{\check{m}}{r}\sin\alpha = k_{\|}\widehat{\mathbf{b}}\cdot\widehat{\rho} + \chi\sin\beta\mathbf{k}\times\widehat{\mathbf{b}}\cdot\widehat{\rho} - \chi\cos\beta\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\cdot\widehat{\rho} \tag{11}$$

$$-\check{k}_{\rho}\|\nabla\rho\|\sin\alpha + \frac{\check{m}}{r} = k_{\|}\widehat{\mathbf{b}}\cdot\widehat{\theta} + \chi\sin\beta\mathbf{k}\times\widehat{\mathbf{b}}\cdot\widehat{\theta} - \chi\cos\beta\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\cdot\widehat{\theta} \tag{12}$$

$$\frac{\check{n}}{R} = k_{\|}\widehat{\mathbf{b}}\cdot\widehat{\phi} + \chi\sin\beta\mathbf{k}\times\widehat{\mathbf{b}}\cdot\widehat{\phi} - \chi\cos\beta\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\cdot\widehat{\phi} \tag{13}$$

where all scalar products must be evaluated using the definitions of $\widehat{\rho}$, $\widehat{\theta}$ and $\widehat{\phi}$ introduced in Ref.[5], taking in particular account that $\hat{\rho}\cdot\hat{\theta} = -\sin\alpha$.

By combining (11) and (12),

$$\begin{aligned}\check{k}_{\rho}\|\nabla\rho\|\cos^2\alpha = {} & k_{\|}\widehat{\mathbf{b}}\cdot\widehat{\rho} + \chi\sin\beta\mathbf{k}\times\widehat{\mathbf{b}}\cdot\widehat{\rho} - \chi\cos\beta\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\cdot\widehat{\rho} \\ & + \left(k_{\|}\widehat{\mathbf{b}}\cdot\widehat{\theta} + \chi\sin\beta\mathbf{k}\times\widehat{\mathbf{b}}\cdot\widehat{\theta} - \chi\cos\beta\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\cdot\widehat{\theta}\right)\sin\alpha\end{aligned} \tag{14}$$

and

$$\begin{aligned}\frac{\check{m}}{r}\cos^2\alpha = {} & \left(k_{\|}\widehat{\mathbf{b}}\cdot\widehat{\rho} + \chi\sin\beta\mathbf{k}\times\widehat{\mathbf{b}}\cdot\widehat{\rho} - \chi\cos\beta\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\cdot\widehat{\rho}\right)\sin\alpha \\ & + k_{\|}\widehat{\mathbf{b}}\cdot\widehat{\theta} + \chi\sin\beta\mathbf{k}\times\widehat{\mathbf{b}}\cdot\widehat{\theta} - \chi\cos\beta\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\cdot\widehat{\theta}\end{aligned} \tag{15}$$

Evaluating $\left(\check{k}_{\rho}, \check{m}, \check{n}\right)$ requires to calculate all scalar products of the forms $\widehat{\mathbf{b}}\cdot\ldots$, $\mathbf{k}\times\widehat{\mathbf{b}}\cdot\ldots$ and $\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\cdot\ldots$ with $\ldots = \widehat{\rho}, \widehat{\theta}, \widehat{\phi}$.

$$\widehat{\mathbf{b}}\cdot\widehat{\rho} = 0 \tag{16}$$

$$\widehat{\mathbf{b}}\cdot\widehat{\theta} = \sigma_I P\cos\alpha \tag{17}$$

$$\widehat{\mathbf{b}}\cdot\widehat{\phi} = \sigma_B T \tag{18}$$

since $\widehat{\phi}\cdot\widehat{\rho}=\hat{\mathbf{s}}\cdot\widehat{\rho}=\widehat{\phi}\cdot\widehat{\theta}=\hat{\mathbf{s}}\cdot\widehat{\phi}=0$ and using the relation $\hat{\mathbf{s}}\cdot\hat{\theta}=\cos\alpha$. Using the vectorial transformations

$$\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\cdot\widehat{\rho}=\mathbf{k}\cdot\left(\widehat{\mathbf{b}}\times\widehat{\rho}\right)=\sigma_B T\mathbf{k}\cdot\hat{\mathbf{s}}-\sigma_I P\mathbf{k}\cdot\widehat{\phi} \tag{19}$$

$$\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\cdot\widehat{\theta}=\mathbf{k}\cdot\left(\widehat{\mathbf{b}}\times\widehat{\theta}\right)=-\sigma_B T\mathbf{k}\cdot\hat{\mathbf{r}}+\sigma_I P\sin\alpha\mathbf{k}\cdot\widehat{\phi} \tag{20}$$

$$\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\cdot\widehat{\phi}=\mathbf{k}\cdot\left(\widehat{\mathbf{b}}\times\widehat{\phi}\right)=\sigma_I P\mathbf{k}\cdot\widehat{\rho} \tag{21}$$

since

$$\begin{aligned}
\widehat{\mathbf{b}}\times\widehat{\rho}&=\sigma_B T\widehat{\phi}\times\widehat{\rho}+\sigma_I P\hat{\mathbf{s}}\times\widehat{\rho}=\sigma_B T\hat{\mathbf{s}}-\sigma_I P\widehat{\phi}\\
\widehat{\mathbf{b}}\times\widehat{\theta}&=\sigma_B T\widehat{\phi}\times\widehat{\theta}+\sigma_I P\hat{\mathbf{s}}\times\widehat{\theta}=-\sigma_B T\hat{\mathbf{r}}+\sigma_I P\sin\alpha\widehat{\phi}\\
\widehat{\mathbf{b}}\times\widehat{\phi}&=\sigma_B T\widehat{\phi}\times\widehat{\phi}+\sigma_I P\hat{\mathbf{s}}\times\widehat{\phi}=\sigma_I P\widehat{\rho}
\end{aligned}$$

and

$$\begin{aligned}
\widehat{\phi}\times\widehat{\rho}&=\hat{\mathbf{s}}\\
\widehat{\phi}\times\widehat{\theta}&=-\hat{\mathbf{r}}\\
\hat{\mathbf{s}}\times\widehat{\rho}&=-\widehat{\phi}\\
\hat{\mathbf{s}}\times\widehat{\theta}&=\sin\alpha\widehat{\phi}\\
\hat{\mathbf{s}}\times\widehat{\phi}&=\widehat{\rho}
\end{aligned}$$

where $\hat{\mathbf{s}}=\sin\alpha\widehat{\mathbf{r}}+\cos\alpha\widehat{\theta}$.

Finally, since

$$\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\cdot\widehat{\rho}=k_{\parallel}\widehat{\mathbf{b}}\cdot\widehat{\rho}-\mathbf{k}\cdot\widehat{\rho}=-\mathbf{k}\cdot\widehat{\rho} \tag{22}$$

$$\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\cdot\widehat{\theta}=k_{\parallel}\widehat{\mathbf{b}}\cdot\widehat{\theta}-\mathbf{k}\cdot\widehat{\theta}=k_{\parallel}\sigma_I P\cos\alpha-\mathbf{k}\cdot\widehat{\theta} \tag{23}$$

$$\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\cdot\widehat{\phi}=k_{\parallel}\widehat{\mathbf{b}}\cdot\widehat{\phi}-\mathbf{k}\cdot\widehat{\phi}=k_{\parallel}\sigma_B T-\mathbf{k}\cdot\widehat{\phi} \tag{24}$$

and

$$\begin{aligned}
\mathbf{k}\cdot\hat{\mathbf{s}}&=\frac{m}{r}\cos\alpha\\
\mathbf{k}\cdot\widehat{\phi}&=\frac{n}{R}\\
\mathbf{k}\cdot\widehat{\mathbf{r}}&=k_\rho\left\|\nabla\rho\right\|\cos\alpha\\
\mathbf{k}\cdot\widehat{\rho}&=k_\rho\left\|\nabla\rho\right\|-\sin\alpha\frac{m}{r}\\
\mathbf{k}\cdot\widehat{\theta}&=-\sin\alpha k_\rho\left\|\nabla\rho\right\|+\frac{m}{r}
\end{aligned}$$

using the fact that $\hat{\rho} \cdot \hat{\mathbf{r}} = \cos\alpha$, the components of $\check{\mathbf{k}}$ are

$$\begin{aligned}\check{k}_\rho \left\|\nabla\rho\right\| \cos^2\alpha = \chi \sin\beta \left(\sigma_B T \frac{m}{r} \cos\alpha - \sigma_I P \frac{n}{R}\right) \\ + \chi \cos\beta k_\rho \left\|\nabla\rho\right\| + k_\parallel \sigma_I P \cos\alpha \sin\alpha \\ + \chi \sin\beta \left(-\sigma_B T k_\rho \left\|\nabla\rho\right\| \cos\alpha + \sigma_I P \sin\alpha \frac{n}{R}\right) \sin\alpha \\ - \chi \cos\beta \left(k_\parallel \sigma_I P \cos\alpha + \sin\alpha k_\rho \left\|\nabla\rho\right\|\right) \sin\alpha \quad (25)\end{aligned}$$

$$\begin{aligned}\frac{\check{m}}{r} \cos\alpha = +k_\parallel \sigma_I P \\ + \chi \sin\beta \sin\alpha \sigma_B T \frac{m}{r} + \chi \cos\beta \cos\alpha \frac{m}{r} \\ - \chi \sin\beta \sigma_B T k_\rho \left\|\nabla\rho\right\| - \chi \cos\beta k_\parallel \sigma_I P \quad (26)\end{aligned}$$

$$\frac{\check{n}}{R} = k_\parallel \sigma_B T + \chi \sigma_I P \sin\beta \left(k_\rho \left\|\nabla\rho\right\| - \sin\alpha \frac{m}{r}\right) - \chi \cos\beta \left(k_\parallel \sigma_B T - \frac{n}{R}\right) \quad (27)$$

By combining (26) and (27), it can be shown, that relation (3) is well satisfied. In Appendix 2, it is also demonstrated that using relations (25), (26) and (27), both relations (6) and (7) are also satisfied, which demonstrate the consistency between the geometrical (explicit) and the physical (implicit) formulation of $\check{\mathbf{k}}$.

References

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2 Appendix

The two first components of $\check{\mathbf{k}}$ may be expressed in a slightly modified form as compared to (25) and (26),

$$\check{k}_\rho\|\nabla\rho\| = \sigma_I P\tan\alpha k_\parallel + \chi\sin\beta\left[\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r} - k_\rho\|\nabla\rho\|\sin\alpha\right) - \sigma_I P\frac{n}{R}\right] -$$

$$-\chi\cos\beta\left(\sigma_I P\tan\alpha k_\parallel - k_\rho\|\nabla\rho\|\right) \tag{28}$$

$$\frac{\check{m}}{r} = \frac{\sigma_I P}{\cos\alpha}k_\parallel + \chi\sin\beta\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\|\right) - \chi\cos\beta\left(\frac{\sigma_I P}{\cos\alpha}k_\parallel - \frac{m}{r}\right) \tag{29}$$

which is more convenient for verifying that the geometrical approach based on the scattering operator $\mathcal{S}_\beta^\chi$ defined by relation (10) is well consistent with the physics of the scattering. We have to check that these transformed components satisfy the conditions (1), (2) and

$$\check{k}_\perp^2 = \check{k_\rho^2}\|\nabla\rho\|^2 - 2\check{k_\rho}\|\nabla\rho\|\frac{\check{m}\sin\alpha}{r} + \frac{\check{m}^2}{r^2} + \frac{\check{n}^2}{R^2} - k_\parallel^2 \tag{30}$$

Using the components of the wave vector $\mathbf{k}$ and$\check{\mathbf{k}}$ before and after the scattering event, these conditions can be written in the form:

$$\sigma_I P\frac{\check{m}}{r}\cos\alpha + \sigma_B T\frac{\check{n}}{R} = \sigma_I P\frac{m}{r}\cos\alpha + \sigma_B T\frac{n}{R} \tag{31}$$

$$k_\rho\check{k}_\rho\|\nabla\rho\|^2 + \frac{m\check{m}}{r^2} + \frac{n\check{n}}{R^2} - \frac{\|\nabla\rho\|\sin\alpha}{r}\left(m\check{k}_\rho + \check{m}k_\rho\right) = k_\perp^2\chi\cos\beta \tag{32}$$

$$\check{k}_\rho^2\|\nabla\rho\|^2 - 2\check{k}_\rho\|\nabla\rho\|\frac{\check{m}\sin\alpha}{r} + \frac{\check{m}^2}{r^2} + \frac{\check{n}^2}{R^2} - k_\parallel^2 = \chi^2 k_\perp^2 \tag{33}$$

2.1 Conservation of $k_\parallel$

$$\check{k}_\parallel = \sigma_I P\frac{\check{m}}{r}\cos\alpha + \sigma_B T\frac{\check{n}}{R} =$$

$$= \sigma_I P\cos\alpha\left[\frac{\sigma_I P}{\cos\alpha}k_\parallel + \chi\sin\beta\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\|\right) - \chi\cos\beta\left(\frac{\sigma_I P}{\cos\alpha}k_\parallel - \frac{m}{r}\right)\right] +$$

$$+\sigma_B T\left[\sigma_B T k_\parallel + \chi\sin\beta\sigma_I P\left(k_\rho\|\nabla\rho\| - \frac{m}{r}\sin\alpha\right) - \chi\cos\beta\left(\sigma_B T k_\parallel - \frac{n}{R}\right)\right] =$$

$$= k_{\parallel}\left(\sigma_I^2 P^2 + \sigma_B^2 T^2\right) + \sigma_I P \sigma_B T \chi \sin\beta \left(\frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\| + k_\rho\|\nabla\rho\| - \frac{m}{r}\sin\alpha\right) -$$

$$-\chi\cos\beta\left[k_{\parallel}\left(\sigma_I^2 P^2 + \sigma_B^2 T^2\right) - \left(\sigma_I P\frac{m}{r}\cos\alpha + \sigma_B T\frac{n}{R}\right)\right] =$$

$$= k_{\parallel} - \chi\cos\beta\left(k_{\parallel} - k_{\parallel}\right) = k_{\parallel}$$

2.2 Direction of $\check{\mathbf{k}}_\perp$

$$\mathbf{k}_\perp \cdot \check{\mathbf{k}}_\perp = k_\rho\check{k}_\rho\|\nabla\rho\|^2 + \frac{m\check{m}}{r^2} + \frac{n\check{n}}{R^2} - \frac{\|\nabla\rho\|\sin\alpha}{r}\left(m\check{k}_\rho + \check{m}k_\rho\right) - k_{\parallel}^2 =$$

$$= k_\rho\|\nabla\rho\|\left\{\sigma_I P\tan\alpha k_{\parallel} + \chi\sin\beta\left[\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r} - k_\rho\|\nabla\rho\|\sin\alpha\right) - \sigma_I P\frac{n}{R}\right]\right\} -$$

$$-k_\rho\|\nabla\rho\|\chi\cos\beta\left(\sigma_I P\tan\alpha k_{\parallel} - k_\rho\|\nabla\rho\|\right) +$$

$$+\frac{m}{r}\left[\frac{\sigma_I P}{\cos\alpha}k_{\parallel} + \chi\sin\beta\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\|\right) - \chi\cos\beta\left(\frac{\sigma_I P}{\cos\alpha}k_{\parallel} - \frac{m}{r}\right)\right] +$$

$$+\frac{n}{R}\left[\sigma_B T k_{\parallel} + \chi\sin\beta\sigma_I P\left(k_\rho\|\nabla\rho\| - \frac{m}{r}\sin\alpha\right) - \chi\cos\beta\left(\sigma_B T k_{\parallel} - \frac{n}{R}\right)\right] -$$

$$-\frac{m}{r}\sin\alpha\left\{\sigma_I P\tan\alpha k_{\parallel} + \chi\sin\beta\left[\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r} - k_\rho\|\nabla\rho\|\sin\alpha\right) - \sigma_I P\frac{n}{R}\right]\right\} +$$

$$+\frac{m}{r}\sin\alpha\chi\cos\beta\left(\sigma_I P\tan\alpha k_{\parallel} - k_\rho\|\nabla\rho\|\right) -$$

$$-k_\rho\|\nabla\rho\|\sin\alpha\left[\frac{\sigma_I P}{\cos\alpha}k_{\parallel} + \chi\sin\beta\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\|\right) - \chi\cos\beta\left(\frac{\sigma_I P}{\cos\alpha}k_{\parallel} - \frac{m}{r}\right)\right] - k_{\parallel}^2 =$$

$$= \left[\left(\sigma_I P\frac{m}{r}\cos\alpha + \sigma_B T\frac{n}{R}\right)k_{\parallel} - k_{\parallel}^2\right] + \chi\cos\beta\left[k_\rho^2\|\nabla\rho\|^2 + \frac{m^2}{r^2} + \frac{n^2}{R^2} - 2k_\rho\|\nabla\rho\|\sin\alpha\right] -$$

$$-\chi\cos\beta k_{\parallel}\left(\sigma_I P\frac{m}{r}\cos\alpha + \sigma T\frac{n}{R}\right) =$$

$$= \left(k_{\parallel}^2 - k_{\parallel}^2\right) + \chi\cos\beta\left(k^2 - k_{\parallel}^2\right) = k_\perp^2\chi\cos\beta$$

2.3 Solution of the dispersion relation

$$\check{k}_\perp^2 = \check{k}_\rho^2 \|\nabla\rho\|^2 - 2\check{k}_\rho \|\nabla\rho\| \frac{\check{m}\sin\alpha}{r} + \frac{\check{m}^2}{r^2} + \frac{\check{n}^2}{R^2} - k_\parallel^2 =$$

$$= \left\{ \sigma_I P \tan\alpha k_\parallel + \chi \sin\beta \left[\frac{\sigma_B T}{\cos\alpha} \left(\frac{m}{r} - k_\rho \|\nabla\rho\| \sin\alpha \right) - \sigma_I P \frac{n}{R} \right] - \chi \cos\beta \left(\sigma_I P \tan\alpha k_\parallel - k_\rho \|\nabla\rho\| \right) \right\}^2 -$$

$$-2\sin\alpha \left[\frac{\sigma_I P}{\cos\alpha} k_\parallel + \chi \sin\beta \frac{\sigma_B T}{\cos\alpha} \left(\frac{m}{r}\sin\alpha - k_\rho \|\nabla\rho\| \right) - \chi\cos\beta \left(\frac{\sigma_B T}{\cos\alpha} k_\parallel - \frac{m}{r} \right) \right] \cdot$$

$$\cdot \left\{ \sigma_I P \tan\alpha k_\parallel + \chi \sin\beta \left[\frac{\sigma_B T}{\cos\alpha} \left(\frac{m}{r} - k_\rho \|\nabla\rho\| \sin\alpha \right) - \sigma_I P \frac{n}{R} \right] - \chi \cos\beta \left(\sigma_I P \tan\alpha k_\parallel - k_\rho \|\nabla\rho\| \right) \right\} +$$

$$+ \left[\frac{\sigma_I P}{\cos\alpha} k_\parallel + \chi \sin\beta \frac{\sigma_B T}{\cos\alpha} \left(\frac{m}{r}\sin\alpha - k_\rho \|\nabla\rho\| \right) - \chi\cos\beta \left(\frac{\sigma_I P}{\cos\alpha} - \frac{m}{r} \right) \right]^2 +$$

$$+ \left[\sigma_B T k_\parallel + \chi \sin\beta \sigma_I P \left(k_\rho \|\nabla\rho\| - \frac{m}{r}\sin\alpha \right) - \chi\cos\beta \left(\sigma_B T k_\parallel - \frac{n}{R} \right) \right]^2 - k_\parallel^2 =$$

$$= \left[\left(\sigma_I^2 P^2 + \sigma_B^2 T^2 \right) k_\parallel^2 - k_\parallel^2 \right] + 2\chi \sin\beta \sigma_I P \sigma_B T k_\parallel \left(k_\rho \|\nabla\rho\| + \frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\| - \frac{m}{r}\sin\alpha \right) -$$
$$-2\chi\cos\beta \left[\left(\sigma_I^2 P^2 + \sigma_B^2 T^2 \right) k_\parallel^2 - \left(\sigma_I P \frac{m}{r}\cos\alpha + \sigma_B T \frac{n}{R} \right) k_\parallel \right] -$$

$$-2\chi^2 \sin\beta\cos\beta \left(\frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\| \right) \left[\frac{\sigma_B T}{\cos\alpha} \left(\frac{m}{r} - k_\rho \|\nabla\rho\| \sin\alpha \right) + \sigma_I P \left(\sigma_B T k_\parallel - \frac{n}{R} \right) \right] +$$

$$+2\chi^2 \sin\beta\cos\beta \left(\frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\| \right) \left[\frac{\sigma_B T}{\cos\alpha} \left(\frac{m}{r} - k_\rho \|\nabla\rho\| \sin\alpha \right) + \sigma_I P \left(\sigma_B T k_\parallel - \frac{n}{R} \right) \right] +$$
$$+\chi^2 \sin^2\beta \left[\left(k_\rho\|\nabla\rho\| - \frac{m}{r}\sin\alpha \right)^2 + \left(\sigma_B T \frac{n}{R}\cos\alpha - \sigma_I P \frac{n}{R} \right)^2 \right] +$$
$$+\chi^2 \cos^2\beta \left(k_\rho^2 \|\nabla\rho\|^2 - 2\frac{m}{r} k_\rho \|\nabla\rho\| \sin\alpha + \frac{m^2}{r^2} + \frac{n^2}{R^2} - k_\parallel^2 \right) =$$
$$= \left(k_\parallel^2 - k_\parallel^2 \right) - 2\chi\cos\beta \left(k_\parallel^2 - k_\parallel^2 \right) + \chi^2 k_\perp^2 \sin^2\beta + \chi^2 \cos^2\beta \left(k^2 - k_\parallel^2 \right) =$$
$$= \chi^2 k_\perp^2 \left(\sin^2\beta + \cos^2\beta \right) = \chi^2 k_\perp^2$$

The expressions for the component of the scattered wave vector derived using the scattering operator satisfy the conservation of $k_\parallel$, the relation between $\mathbf{k}_\perp$ and $\check{\mathbf{k}}_\perp$ and the dispersion relation.